

Raj Ankit

Department of Physics and Material Science, Thapar Institute of Engineering and Technology,
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Work experience

Lecturer (Contractual)

July 2022 – June 2023

Department of Physics and Materials Science, Thapar Institute of Engineering and Technology,
Patiala, Punjab, India.

- Taking an undergraduate lab course and
- Tutorials for the Engineering Materials course.
- Assisted in writing a research proposal to obtain funding from various agencies.

Assistant Professor (Contractual)

July 2023 – June 2024

Department of Physics and Materials Science, Thapar Institute of Engineering and Technology,
Patiala, Punjab, India.

- Taking an undergraduate lab course.
- Demonstrate the instrument capability with real-time analysis to various undergraduate and postgraduate students across the department.
- Tutorials for the Engineering Materials course.

Assistant Professor (Contractual)

July 2024 – Sept. 2024

- Taking an undergraduate lab course.
- Tutorials for the Applied Physics course

SRF

Oct. 2024 – Dec. 2025

- Synthesis and characterization of ceramics for a thermistor.
- Successfully developed the PTCR thermistor for industrial applications.
- Fabricated roll milling for homogeneous powder mixing.
- Fabricated a sample holder for high-temperature dielectric measurement.
- Electroplating and electroless plating of nickel on ceramic pellets.

Assistant Professor (Contractual)

July 2024 – Sept. 2024

- Taking an undergraduate lab course.
- Tutorials for the Applied Physics course

Education

Ph.D. Physics

Jan 2022 - Present

CGPA 8.85

Department of Physics and Materials Science, Thapar Institute of Engineering and Technology,
Patiala, Punjab, India.

M. Sc. Physics

July 2018 – July 2020

CGPA 9.01

Department of Physics and Materials Science, Thapar Institute of Engineering and Technology,
Patiala, Punjab, India.

B. Sc. Physics

July 2014 – May 2017

86.38%

Tilkamanjhi Bhagalpur University, Bhagalpur, Bihar, India.

Thesis work

Ph.D. Thesis: *Biopolymers-based triboelectric nanogenerator energy harvesting and sensor applications.*

Brief: A systematic approach was adopted to study the triboelectric properties of biopolymer-based films. Various methods have been adopted for synthesis.

1. Multiple biodegradable polymers, including **chitosan, starch, silk, and cellulose**, were systematically explored to understand the influence of molecular structure, functional groups, and surface polarity on triboelectric charge generation.
2. **Chitosan**-based systems demonstrated strong positive triboelectric behavior due to abundant amino and hydroxyl groups, enabling efficient charge transfer during contact electrification.
3. **Cellulose and cellulose-derived structures** were investigated owing to their hierarchical fibrous morphology, high surface area, and excellent biodegradability, which enhanced contact efficiency and triboelectric performance.
4. **Micro/nano structuring** through **electrospinning** and **solution casting** significantly increased the effective contact area, resulting in improved electrical output characteristics.
5. Dielectric enhancement was achieved by incorporating ceramic fillers such as **BaTiO₃**, improving polarization, charge trapping, and energy storage capability of the biopolymer matrices.
6. Device-level demonstrations included **self-powered speed sensing, vibration/acoustic energy harvesting, capacitor charging, a game controller, and a toy car controller using TENGs**, highlighting multifunctional sensing capability.

M.Sc. Thesis: *Energy harvesting from lead-free piezoelectric material.*

Brief: The research focused on the development of **environmentally benign piezoelectric energy-harvesting systems using lead-free materials as alternatives to conventional lead-based ceramics**, such as PZT. BCZT powders were synthesized and structurally analyzed to confirm phase purity and to improve flexibility and device applicability. Polyvinylidene **fluoride (PVDF)** polymer was incorporated as a piezoelectric matrix owing to its lightweight nature, mechanical flexibility, and ability to exhibit β -phase polarization.

Awards

- **1st Place** in district-level science quiz in classes 11 and 12 conducted by Vidya-Bharti.
- **Distinction** in the **B.Sc. (Physics)** examination held by Tilkamanjhi Bhagalpur University, Bhagalpur, India.
- **Top 5 rank holder** in the **M. Sc. (Physics)** Thapar Institute of Engineering and Technology, Patiala, Punjab, India.
- **1st Place poster award** in the International Conference on Research Advancements and Challenges in Glass and Ceramics (RAICGC -2025), held during November 27-29, 2025, at the 89th annual session of the Indian Ceramic Society, Indian Institute of Technology Bombay, Mumbai, INDIA.
- **Senior Research Fellowship (SRF)** in an industry-sponsored project by AJM India Pvt. Ltd.
- **Article Processing Charge (APC) waiver** awarded by *Materials Today Advances* (Elsevier) in recognition of research quality.

Research interest

- Triboelectric Nanogenerators (TENGs) and Self-Powered Systems
- Energy Harvesting from Mechanical and Acoustic Sources
- Biopolymer-Based Functional Materials for Sustainable Electronics

- Polymer–Ceramic Nanocomposites (BCZT, BaTiO₃, PVDF Systems)
- Self-Powered Sensing and Smart Monitoring Systems
- Surface Engineering and Micro/Nanostructuring for Charge Enhancement
- Dielectric and Ferroelectric Materials for Energy Applications
- Waste-Derived Functional Materials and Circular Economy Materials
- Electrospinning and Solution-Processed Functional Films
- Hybrid Energy Harvesting Devices (Piezoelectric–Triboelectric Coupling)
- Electroceramics and Functional Ceramic Devices
- Thermistor Materials and Temperature Sensors for Industrial Applications

Facilities developed

- Designed and installed a sample holder for temperature-dependent dielectric measurement.
- Designed and installed a linear-motor-based tapping device for triboelectric measurements.
- Fabricated roll milling for homogeneous mixing of powder via a solid-state reaction route.

Technical skills

Characterization Skills

- Scanning electron microscopy (SEM), XRD, FESEM, EDX, Raman, FTIR, dielectric, impedance, piezoelectric, triboelectric measurements.
- Current-Voltage (I-V) and Capacitance-Voltage (C-V) measurements.

Other Skills

- Synthesis of nanopowder by organic route (Sol-gel and solid-state reaction route) and well-versed with the powder compaction technique.
- Fabrication of thick ceramic films by tape casting, electrospinning, spray coating, and reactive template grain growth (RTGG).
- Fabrication of thick polymer films by tape casting, electrospinning, and spray coating.
- Electroplating and electroless plating of nickel on ceramic.
- Proficiency in software such as OriginPro, X'Pert Highscore, ImageJ, and MS Office.

Publications

Journal Publication

1. **R Ankit**, P Prakash, R Singla, J Kolte, "Energy from trash: a flexible, facile, and robust triboelectric nanogenerator based on waste polystyrene and application as a human–machine interface" **RSC Adv.**, (2026) 16, 4107-4115.
2. **Raj Ankit**, Simranjeet Kaur, Shinar Athwal, Taranveer Kaur, and Jayant Kolte, "An edible microbial cellulose-based triboelectric nanogenerator: a sustainable approach for energy harvesting" **Materials Advances** (2026). 10.1039/D5MA01186H.
3. **Raj Ankit**, Jayant Kolte, "High-performance piezoelectric 2D-single crystalline BaTiO₃ /chitosan composite-based triboelectric nanogenerator for vibrational energy harvesting and game controller" **Materials Today Advances** (2025) 28, 100630.
4. **Raj Ankit**, Jayant Kolte, "Engineering of Chitosan/poly (vinyl alcohol) nanofibrous membrane as triboelectric nanogenerator for energy harvesting and self-powered speed sensor" **Journal of Materials Science: Materials in Electronics** (2025) 36 (22) 1366.
5. **Raj Ankit**, Tirthankar Chakraborty, Jayant Kolte, "Scalable Electrospun Chitosan/PVA Nanofibers: A Pathway for Triboelectric Nanogenerator-based Energy Harvesting", **Polymer** (2025) 335, 128848.

Conference Presentation

1. Raj Ankit, Jayant Kolte "Highly sensitive triboelectric nanogenerator based on Chitosan/PVA for energy harvesting and human motion sensors" International conference on recent advances in biofuels and biomaterials (ICRABB -2023)" held during October 13-14, 2023, in Department of

- Chemical Engineering, Dr B R Ambedkar National Institute of Technology, Jalandhar, Punjab (INDIA).
2. Raj Ankit, Jayant Kolte "Highly sensitive triboelectric nanogenerator based on Chitosan/PVA for energy harvesting and human motion sensors" International conference on sustainable energy and environment (ICSEE -2024)" held during February 23-24, 2024, in Department of Chemical Engineering, Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh (INDIA).
 3. Raj Ankit, Rahul Kumar, Sachet Shah, Vineet Shah, Jayant Kolte "Tuning of electrical properties of barium titanate-based ceramics for the semiconducting industry" International conference on research advancements and challenges in glass and ceramics (RAICGC -2025)" held during November 27-29, 2025, at 89th annual session of Indian ceramic society, Indian Institute of Technology Bombay, Mumbai, INDIA.

References

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